

INSTRUCTION

Tamil AI Learning Assistant

Project Setup & Execution Guide

Prerequisites

Ensure the following are installed on your system:

- * Java (JDK 17 or above)
- * Python (3.9 or above)
- * Flutter SDK
- * XAMPP (for MySQL Server)
- * Ollama (for local AI model)
- * Maven

Project Structure

After extracting the ZIP, the folder structure will be:

Tamil-AI-Learning-Assistant/

```
|  
├── backend/  
├── ai-service/  
├── frontend/  
├── README.md  
└── demo.txt
```

Step 1: Database Setup (XAMPP)

1. Open ****XAMPP Control Panel****

2. **Start:**

* Apache

* MySQL

3. Open browser:

`http://localhost/phpmyadmin`

4. **Navigate :**

`C:\xampp\mysql\bin`

Open CMD this location

Command:

`mysql -u root -p`

5. Then run:

6. `CREATE DATABASE tamil_ai;`

7. `SHOW DATABASES;`

8. Configure database connection:

Open file:

`backend/src/main/resources/application.properties`

Update:(if any changes in Mysql username and password)

`spring.datasource.url=jdbc:mysql://localhost:3306/tamil_ai`

`spring.datasource.username=root`

`spring.datasource.password=`

`spring.jpa.hibernate.ddl-auto=update`

Step 2: Run Spring Boot Backend

1. **Navigate (CMD):**

`cd backend`

2. **Run:**

`mvnw.cmd spring-boot:run`

3. Backend runs at:

`http://localhost:8080`

Ensure backend starts without errors

Step 3: Run AI Service (FastAPI)

1. Navigate (CMD):

`cd ai-service`

2. Install dependencies:

`pip install -r requirements.txt`

3. Run:

`uvicorn main:app --reload`

4. Runs at:

<http://localhost:8000>

Step 4: Run Ollama Model

1. Install Ollama (Download and Install ollama)

2. Run model:

`ollama run gemma4:e2b`

3. Verify:

`http://localhost:11434`

Keep running in background

Step 5: Flutter Setup

Step 5.1: Configure API

Open:

`frontend/lib/services/api_service.dart`

Mobile Testing (Recommended)

`http://<YOUR_LAPTOP_IP>:8080/api`

Find Your IP (Windows)

Run(CMD):

ipconfig

Find:

IPv4 Address

Example:

192.168.1.5

Final API:

`http://192.168.1.5:8080/api`

Notes:

- * Same Wi-Fi required
- * Do NOT use localhost on mobile

Step 5.2: USB Debugging

1. Enable Developer Options
2. Enable USB Debugging
3. Connect device via USB
4. Verify:(Show your Device Name)
flutter devices

Step 5.3: Run Flutter App (CMD)

1. Navigate:

```
cd frontend
```

2. Install:

```
flutter pub get
```

3. Run:

```
flutter run
```

Authentication

- * Register user
- * Login
- * JWT stored locally
- * Auto-login enabled

Features

- * Tanglish → Tamil Translation
- * Tamil Sentence Rephrasing
- * Tamil Sentence Generation

Important Notes

- * Ensure all are running:
- * Backend
- * AI Service
- * Ollama
- * AI may be slow (CPU-based)

Troubleshooting

Issue	Solution
Backend not running	Check DB config
AI not working	Run FastAPI + Ollama
Device not detected	Enable USB debugging
Mobile not connecting	Use correct IP
Slow response	Expected

Execution Order

1. Start XAMPP
2. Run Backend (`backend`)
3. Run AI Service (`ai-service`)
4. Run Ollama
5. Configure API
6. Run Flutter (`frontend`)

Project Overview

Full-stack AI-powered Tamil Learning System:

- * Flutter (Frontend)
- * Spring Boot (Backend)
- * FastAPI (AI Layer)
- * Ollama (LLM)
- * MySQL (Database)

Final Note

All services must run simultaneously:

****Flutter → Backend → AI Service → Ollama → Response****